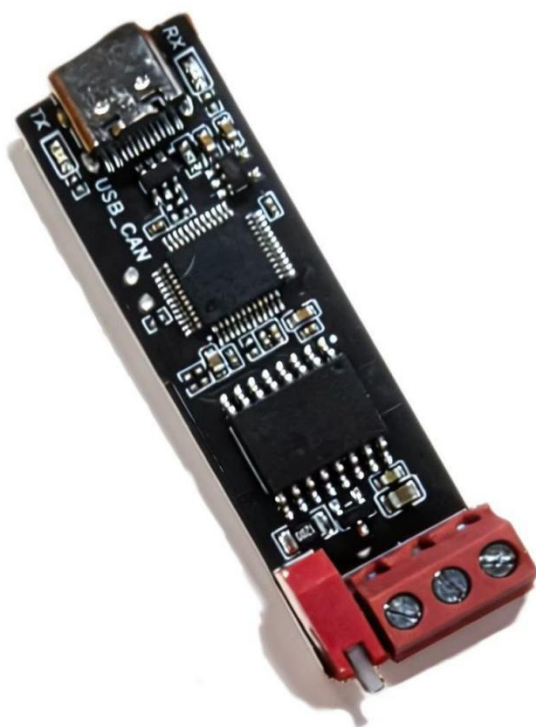


CANable 用户手册

CANable User Manual

V1.0



版本变更说明

Version Change Description

UART 通信接口用户手册版本变更说明				
UART Communication Interface User Manual Version Change Notes				
序号	版本号	变更日期	修订内容	变更人
1	v1.0	20260305	初版发布 First Edition Released	--

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一、产品介绍

product introduction

1.1 产品功能介绍

Product Features

CANable 是我司关节系列产品的专用通信接口，主要用于实现关节模块与公司自研上位机软件 EYouCanOpenTool(PHU) 之间的通信连接。

CANable is a dedicated communication interface for our joint series products, designed to enable seamless communication between joint modules and the company's proprietary EYouCanOpenTool (PHU) software.

EYouServoStudio 是一款专为关节产品设计的串口调试与控制软件。该软件通过 CANopen 通信协议与谐波关节硬件建立连接，为用户提供以下核心功能：

EYouServoStudio is a serial debugging and control software specifically designed for articulated products. It connects to harmonic joint hardware via the CANopen communication protocol, providing users with the following core functionalities:

①核心参数监控与状态显示：能够实时连接并识别关节产品读取并显示其基本运行参数。

Core parameter monitoring and status display: Capable of real-time connection and recognition of joint products to read and display their basic operational parameters

②多样化的运动模式控制：支持发送标准的 CANopen 运动控制指令，包括轮廓位置、轮廓速度、轮廓力矩、同步位置、同步速度、同步力矩等等运动模式。

Diverse motion control modes: Supports sending standard CANopen motion control commands, including profile position, profile speed, profile torque, synchronized position, synchronized speed, synchronized torque, and other motion modes.

③全面的 SDO 读写服务：具备完整的服务数据对象（SDO）访问功能，可用于对设备的对象字典进行读写操作。

Comprehensive SDO read/write service: Provides full access to service data objects (SDO), enabling read and write operations on the device's object dictionary.

④状态可视化：通过运动曲线实时反映关节的运动状态，帮助用户更直观地进行调试与开发。

Status visualization: Real-time motion curves reflect joint movement status, helping users debug and develop more intuitively.

1.2 性能说明

Performance description

①PC 接口符合 USB2.0 全速规范，兼容 USB3.0；

The PC interface complies with the USB2.0 full-speed specification and is compatible with USB3.0

②支持 CAN2.0A 帧格式，符合 ISO/DIS11898 规范；

It supports CAN2.0A frame format and complies with ISO/DIS11898 specifications

③使用 USB 总线电源供电 (DC+5V, 150mA)；

Powered by USB bus power supply (DC+5V, 150mA);

④上位机支持 Win7、Win8、Win10、Win11 等 Windows 操作系统；

The host computer supports Win10, Win11 and other Windows operating systems;

⑤工作温度：-20°C~+70°C；工作湿度：15%~90%RH，无凝露。

Working temperature: -20°C~+70°C; Working humidity: 15%~90%RH, no condensation.

⑥产品附带一个 120Ω 的电阻，可通过拨码打开。

The product includes a 120Ω resistor that can be activated via a rotary dial switch.

二、产品使用方法介绍

Product Usage Instructions

2.1 产品外形展示

Product Appearance

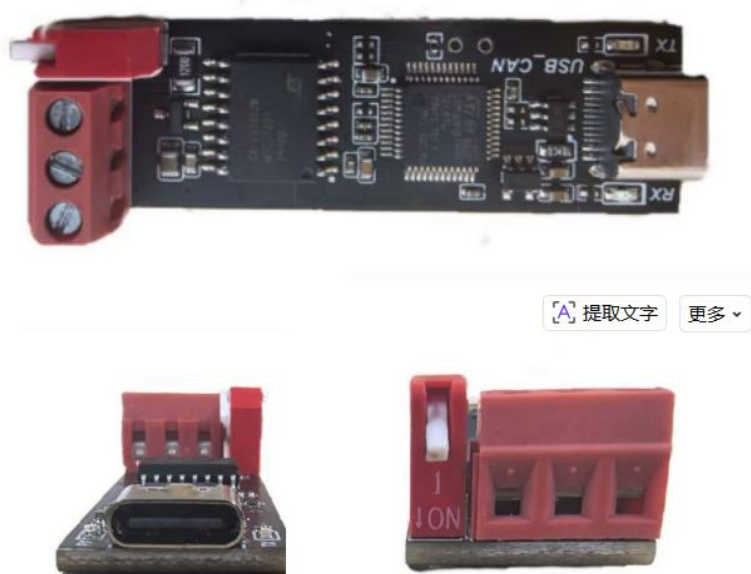


图 2-1 产品外形展示

Figure 2-1 Product Appearance Display

说明：产品长约 55mm，宽约 17mm（由于公差存在少许误差是正常的）

Note: The product measures approximately 55mm in length and 17mm in width (minor variations due to tolerances are normal).

2.2 产品引脚定义

Product Pin Definition

按上图俯视图的方向，左侧为 CANopen 接线端口，右侧为 USB 接口。
CANopen 接线端口见上图外观展示图，最左侧为 120 欧姆电阻拨码开关，三个接口从左到右分别是 CAN_H, CAN_L, GND。

As shown in the top view, the CANopen port is on the left and the USB port on the right. The CANopen port is depicted in the exterior diagram above, featuring a

120-ohm potentiometer switch on the far left. The three ports, from left to right, are CAN_H, CAN_L, and GND.

2.3 产品接线方式

product wiring method

USB 接口端连接电脑，CANopen 接口端连接关节的 CAN 通信口，如图所示：注意关节接线处最右侧的蓝线与 CANable 接口端最左侧相连，中间的白色线与 CANable 中间线相连，GND 不连线，引脚在产品的背面也有具体的标注。

Connect the USB interface to the computer and the CANopen interface to the joint's CAN communication port, as shown in the diagram. Note that the blue wire on the far right of the joint's wiring connects to the far left of the CANable interface, while the white wire in the middle connects to the CANable center line. The GND wire is not connected, and specific markings are also provided on the product's back.

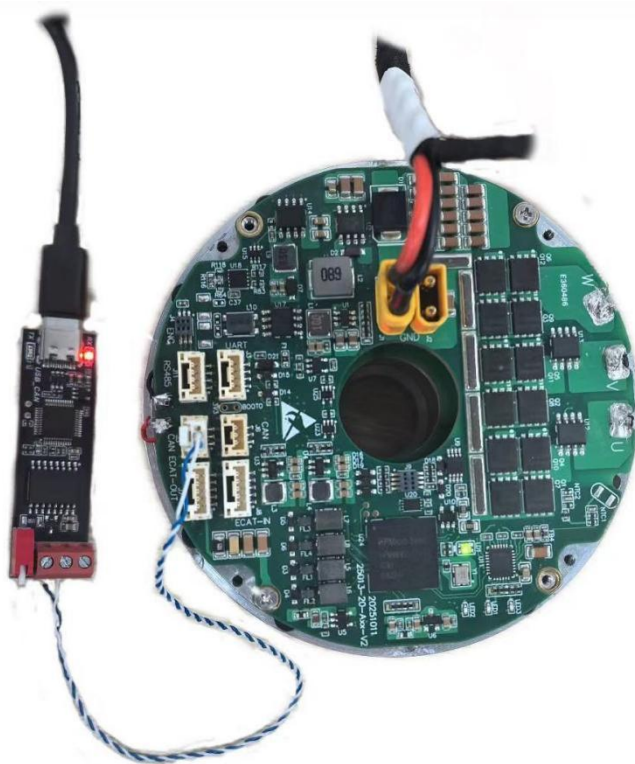


图 2-2 产品接线图

Figure 2-2 Product Wiring Diagram

三、上位机的使用

The use of the host computer

3.1 上位机的下载步骤

Download steps for the host computer

EYouCanOpenTool(PHU)的下载步骤如下，点击软件包进行下载，下载完毕后点击信任此设备，如果在网页无法下载，需要保留文件再下载，详细的操作过程如下：

To download EYouCanOpenTool (PHU), click the software package. After downloading, click Trust This Device. If the download fails on the web, save the file and try again. The detailed steps are as follows:

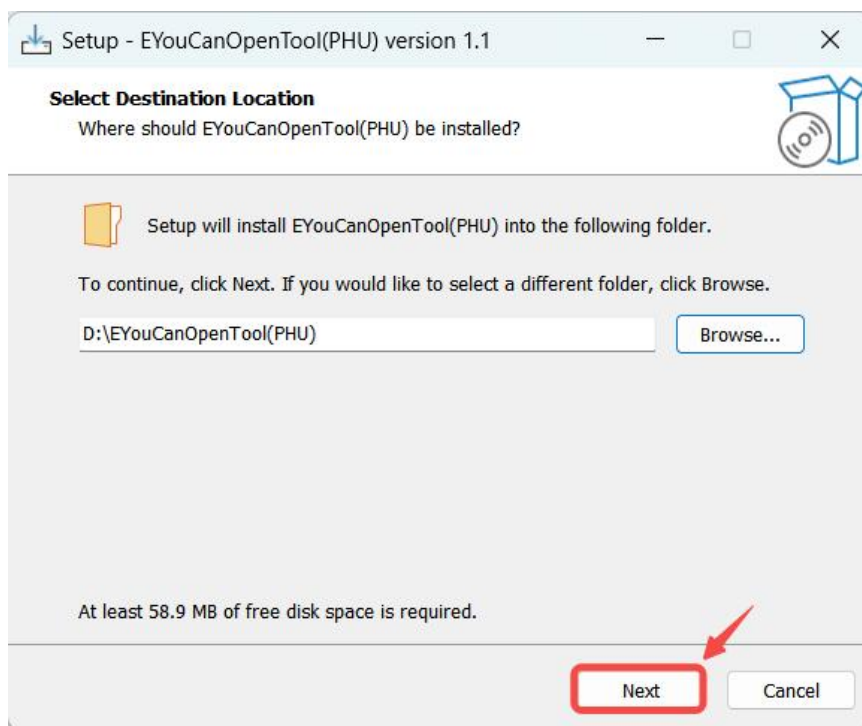


图 3-1 上位机的下载步骤 1

Figure 3-1 Step 1 of the PC Download

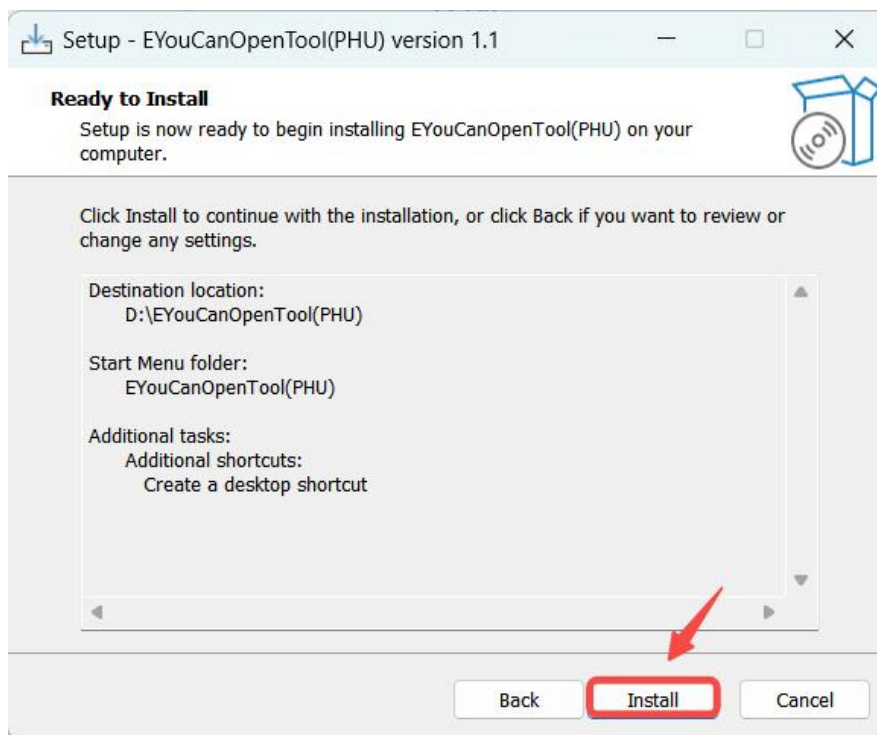


图 3-2 上位机的下载步骤 2

Step 2 of the host computer download

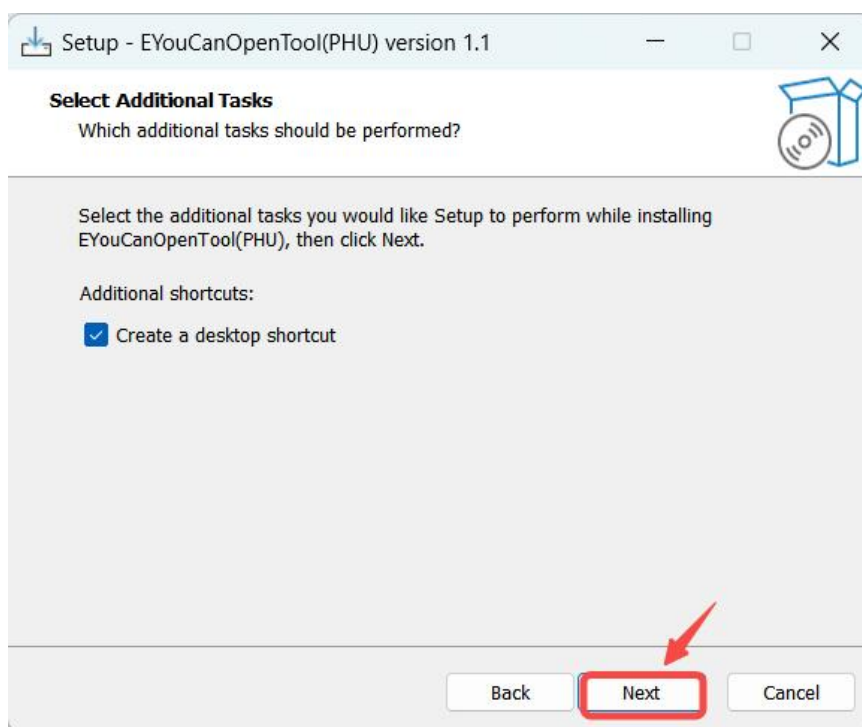


图 3-3 上位机的下载步骤 3

Figure 3-3: Step 3 of the PC Download

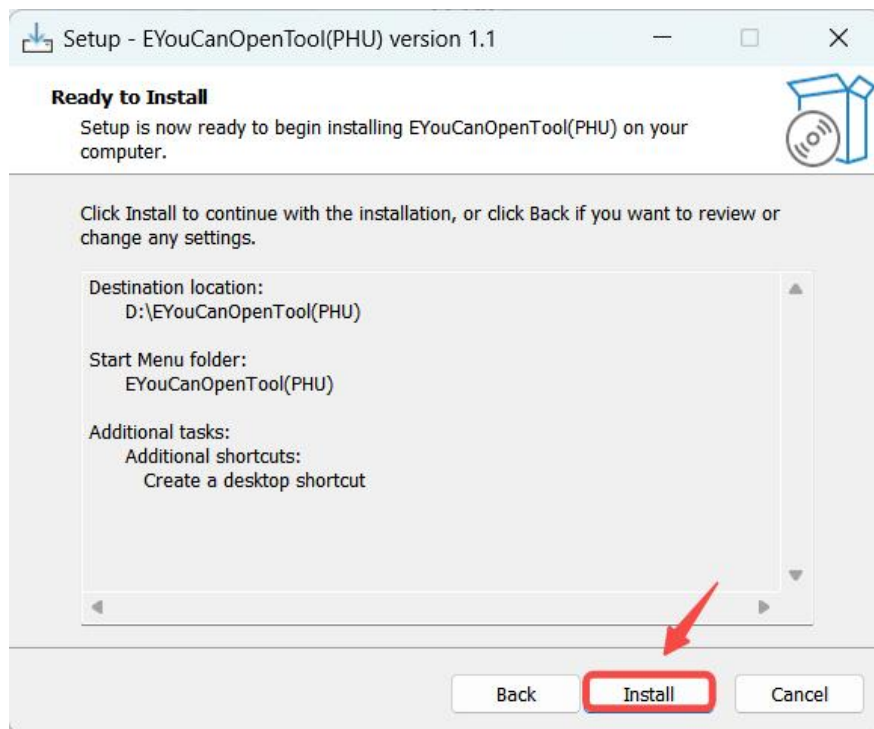


图 3-4 上位机的下载步骤 4

Figure 3-4: Step 4 of the PC Download

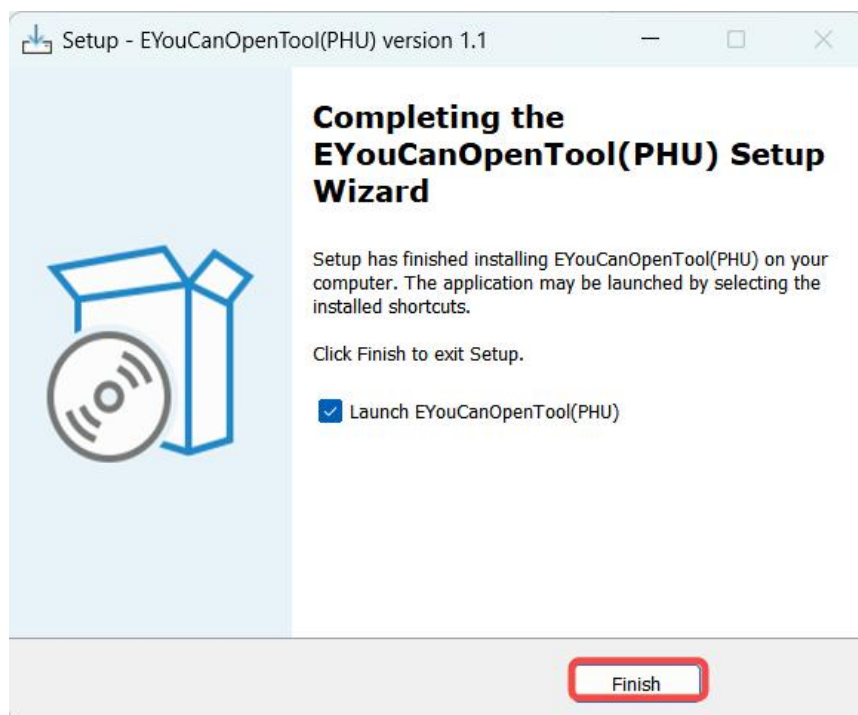


图 3-5 上位机的下载步骤 5

Figure 3-5: Step 5 of the PC Download

3.2 上位机的连接方法

Connection Method of Host Computer

连好线后先打开电源，打开上位机，在初始界面选择 Canable，成功打开后产品的 RX 端口会显现红色，TX 端口会显现绿色，两者会不断闪烁，在上位机上也会显示出产品的相关参数，如图所示：

After connecting the cable, power on the device and boot the host computer. On the initial screen, select 'Canable'. Upon successful activation, the product's RX port will light up red while the TX port glows green, with both ports flashing continuously. The host computer will also display the product's relevant parameters, as shown in the figure.



图 3-6 上位机的连接步骤 1

Figure 3-6: Step 1 of the Host Computer Connection

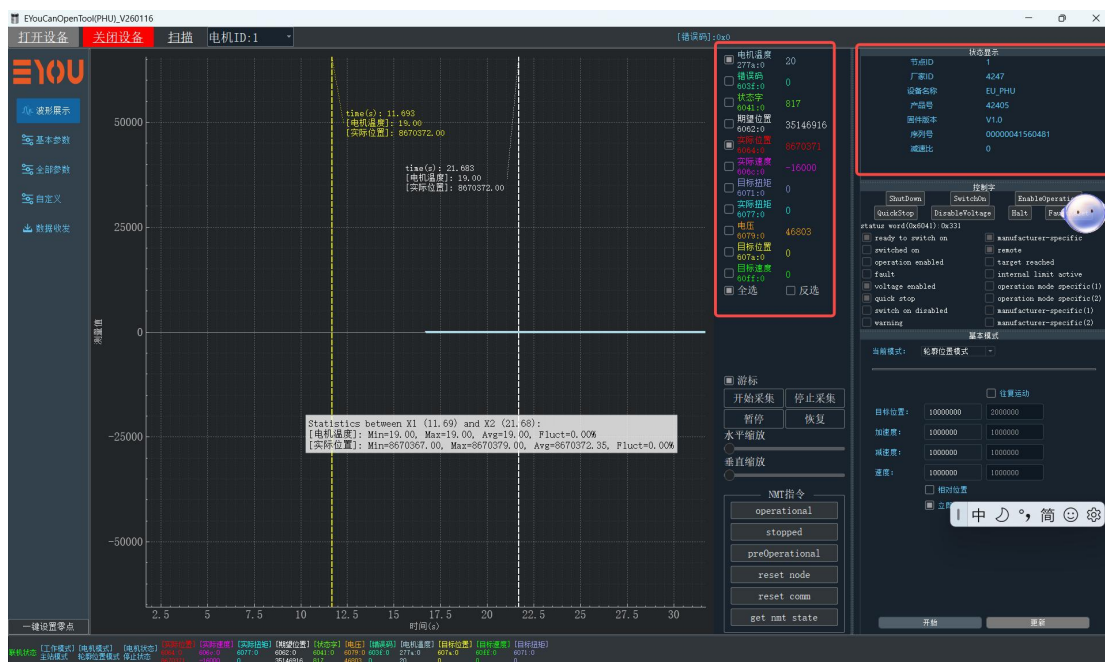


图 3-7 上位机的连接步骤 2

Figure 3-7: Step 2 of the Host Computer Connection